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Abstract

This project presents the complete factory layout design and simulation of Yuwan-Sara Tool Factory, a precision CNC machining facility developed using DELMIA software on the Dassault Systemes 3DEXPERIENCE platform. The primary focus of the project is space optimisation, workflow efficiency, and production planning within a 16 m × 12 m factory floor area.

The factory is designed to accommodate three Vertical Machining Centers (VMC MC 01, VMC MC 02, VMC MC 03), one dedicated inspection room, a raw material storage zone, a finished goods storage zone, a fixture and tool preparation area, and a restroom for personnel welfare. The facility is staffed by 16 skilled personnel organised across four functional roles ensuring uninterrupted production flow from raw material entry to finished goods dispatch.

The facility is capable of manufacturing a wide range of precision engine components including piston heads, connecting rods, and crankshafts — components that demand tight dimensional tolerances and multi-stage machining sequences. The complete layout was designed and simulated in DELMIA to achieve an efficient workflow, resulting in significant improvements in space utilisation (+21%), machine utilisation (+43%), and a reduction in material handling time (−57%) and dead space (−75%) compared to an unoptimised baseline arrangement.

This report is organised into six chapters covering the introduction and company background, plant layout design in DELMIA, factory flow simulation, prototype model documentation, conclusion, and references.

1. Introduction

In modern manufacturing industries, efficient factory planning and process optimisation play a major role in improving productivity, reducing operational costs, and ensuring smooth workflow. DELMIA, developed by Dassault Systemes, is an advanced digital manufacturing and simulation software widely used for factory layout planning, workflow simulation, robotic operations, ergonomics, and production system analysis. It enables engineers to create a virtual model of the factory environment — including machines, workstations, workers, material storage areas, and movement paths — before the actual physical setup is implemented.

Factory flow planning is an important aspect of industrial engineering because it defines how raw materials move through different stages of production such as machining, inspection, assembly, and final storage. A well-designed factory flow reduces unnecessary movement, minimises material handling time, avoids congestion, improves worker efficiency, and ensures better utilisation of available floor space. Proper arrangement of CNC machines, inspection zones, and storage locations also supports continuous production with minimum delay.

The use of simulation before real-time implementation is highly beneficial because it allows engineers to identify design problems, workflow interruptions, machine placement issues, and safety concerns in a risk-free virtual environment. Instead of changing the layout after installation, the complete system can be tested, modified, and optimised digitally. This reduces the chances of costly errors and helps in selecting the most efficient production arrangement.

Another major advantage of DELMIA simulation is the reduction in cost and time. Since the production line is validated virtually, industries can reduce machine relocation costs, lower setup time, improve manpower allocation, and shorten project development cycles. It also supports better decision-making by providing a clear visualisation of worker movement, machine utilisation, and material flow, ultimately leading to higher productivity and better space optimisation.

About the Company

Yuvan-Sara Tool Factory is a precision CNC machining facility specialising in the manufacture of custom-engineered and batch-production metal components for the automotive, mechanical, and general engineering sectors. The factory was established with the objective of providing high-quality machined parts to both domestic and export markets, with a strong emphasis on dimensional accuracy, surface quality, and on-time delivery.

The company operates from a purpose-designed facility with a gross floor area of 16 m × 12 m (192 sq m), housing three state-of-the-art Vertical Machining Centers, a dedicated quality inspection room, raw material and finished goods storage zones, and a fixture preparation area. The facility is staffed by 16 trained personnel across four functional roles: raw material handling, CNC machine operation, quality inspection, and finished goods handling.

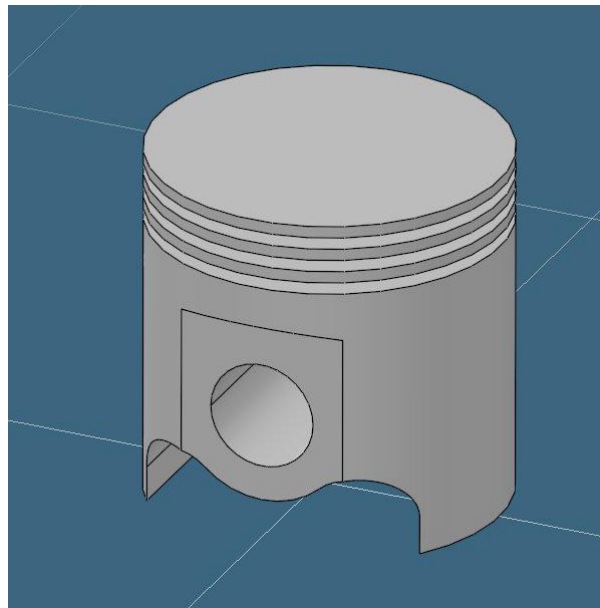
Yuvan-Sara Tool Factory follows a customer-driven production philosophy, accepting both one-off custom part orders to customer-supplied drawings and medium-volume batch orders of standardised components. The company is committed to continuous improvement through digital manufacturing tools, with the factory layout and workflow designed and validated using DELMIA on the Dassault Systemes 3DEXPERIENCE platform — ensuring that every production decision is supported by data-driven simulation before physical implementation.

Parameter	Details
Company Name	Yuvan-Sara Tool Factory
Facility Area	16 m × 12 m (192 sq m Gross Floor Area)
Production Type	Custom Parts & Batch Production
Industry Served	Automotive, Mechanical, General Engineering
CNC Machines	3 × Vertical Machining Centers (3-Axis, ATC)
Total Workforce	16 Personnel (4 roles)
Quality Control	In-process and final inspection on every part
Design Software	DELMIA — 3DEXPERIENCE Platform, Dassault Systemes
2D CAD	Autodesk AutoCAD (DXF import to DELMIA)

Manufacturing Products

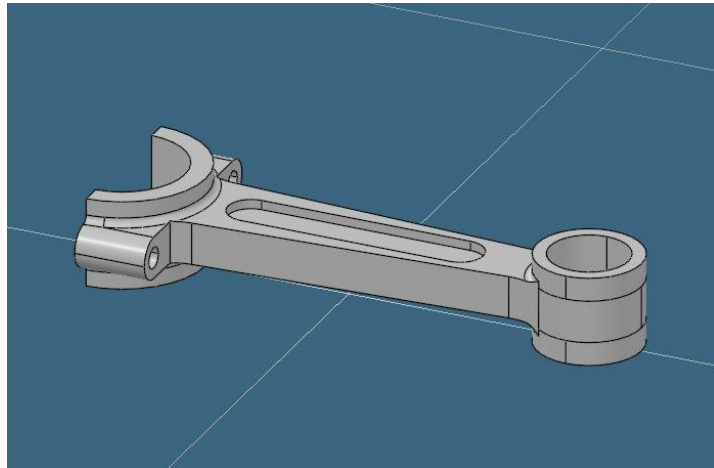
Yuvan-Sara Tool Factory is equipped and organised to manufacture a wide range of precision machined components. The three VMC machines, combined with the structured inspection and handling workflow, provide the capability to produce components of varying complexity — from simple prismatic blocks and flanges to geometrically complex engine components. The facility specialises in the following representative product families:

(a) Piston Head



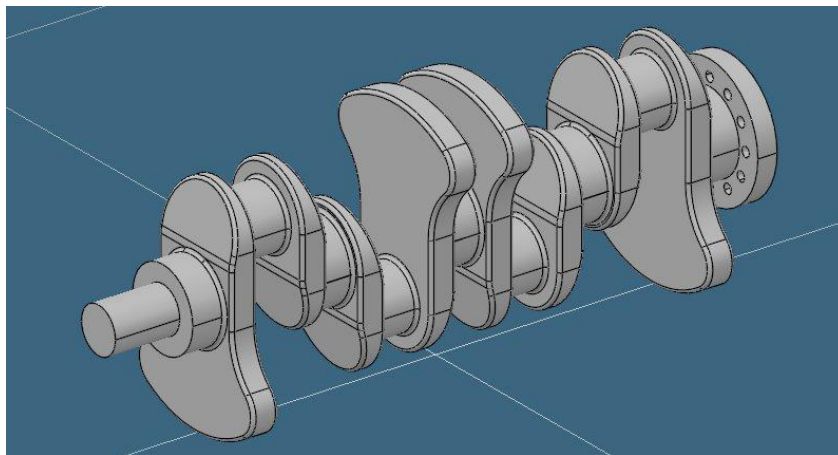
The piston head is a cylindrical engine component machined from aluminium alloy or cast iron billet. The manufacturing sequence involves face milling of the crown surface (VMC MC 01), ring groove profiling (VMC MC 02), and precision bore machining of the gudgeon pin hole to h6 tolerance (VMC MC 03). Critical tolerance requirements include the piston outer diameter for correct bore clearance, ring groove width and depth, and gudgeon pin bore diameter. The multi-operation nature of piston head machining makes the parallel VMC configuration at Yuvan-Sara particularly suitable, as different operations can be distributed across machines to balance cycle times and reduce total throughput time.

(b) Connecting Rod



The connecting rod transmits the reciprocating motion of the piston to the rotational motion of the crankshaft. It is one of the most geometrically complex components produced at the facility, requiring milling of the I-beam shank profile (VMC MC 01), precision boring of the big-end and small-end bores to H7 tolerance (VMC MC 02), drilling and tapping of the cap bolt holes, and precision surface finishing (VMC MC 03). The dedicated fixture area supports the multi-face fixturing setups required for this component. The oil slot visible in the 3D model adds a contouring operation to the machining sequence, demonstrating the 2.5D capability of the VMC fleet.

(b) Crankshaft



The crankshaft is the most complex representative product, requiring the highest number of machining operations and the tightest geometric tolerances in the engine assembly. The manufacturing process involves rough profiling of the cheek and counterweight geometries (VMC MC 01), precision milling of the main journal and crankpin diameters (VMC MC 02), and finishing of the oil holes, keyway features, and the flange bolt circle (VMC MC 03). The crankshaft demands particularly rigorous inspection after each major machining stage, making the dedicated inspection room and inspection team essential to maintaining conformance throughout the multi-stage production sequence. Given its complexity, the crankshaft represents the design ceiling of the facility's current capability and serves as a benchmark for VMC performance qualification.

Operations Used to Manufacture

The manufacturing operations carried out at Yuvan-Sara Tool Factory are entirely CNC-based, executed on the three Vertical Machining Centers. The operations span the full range of milling, drilling, and boring capabilities available on 3-axis VMC platforms, from aggressive roughing passes to precision finishing operations. Table 1.2 below provides a comprehensive listing of operations used across the product range, including the assigned VMC machine, tooling used, and the feature produced.

Operation	VMC Machine	Tool Used	Output / Feature
Face Milling	VMC MC 01	Face Mill Cutter	Flat reference surface on crown/face
Profile Milling	VMC MC 01	End Mill Cutter	External profiles, shoulders, ribs
Pocket Milling	VMC MC 01	End Mill Cutter	Internal recesses, cavities, windows
Drilling	VMC MC 02	Twist Drill / Carbide Drill	Holes for pins, bolts, oil ways
Boring	VMC MC 02	Boring Bar	Precision diameter bores to H7/H6
Tapping	VMC MC 02	Tap Tool	Internal metric/UNC threads
Contouring	VMC MC 02	Ball-nose End Mill	Curved, complex 2.5D profiles
Finishing Passes	VMC MC 03	Fine End Mill	Tight-tolerance final surfaces
2.5D Contouring	VMC MC 03	Ball-nose Mill	Final complex surface geometry
Engraving	VMC MC 03	Engraving Cutter	Part marking and identification

All operations are executed under CNC program control using G-code routines generated by CAM software. Cutting parameters including spindle speed, feed rate, depth of cut, and coolant application are pre-defined for each operation and stored in the machine program memory, ensuring repeatability and consistency across production batches. The Automatic Tool Changer (ATC) on each VMC enables unattended tool selection between operations, minimising cycle interruptions and improving machine utilisation significantly compared to manual tool changing.

The distribution of operations across three machines is a deliberate design decision. VMC MC 01 handles all high material removal rate roughing operations that generate the most heat, vibration, and cutting force — protecting the precision spindle of VMC MC 03 from the stresses of roughing work. VMC MC 02 manages the diverse drilling, boring, and threading operations that require a large ATC carousel (24 tools) to accommodate the variety of tooling needed for high-mix production. VMC MC 03 operates at the highest

spindle speed and lightest cutting parameters, maintaining the geometric stability needed for tight-tolerance finishing surfaces.

Equipment Used

The Yuvan-Sara Tool Factory is equipped with a carefully selected set of production machinery, measurement instruments, and material handling equipment, each chosen for its contribution to the facility's precision machining capability and workflow efficiency. Table 1.3 provides a complete listing of the principal equipment items and their roles within the production system.

Equipment	Specification	Quantity	Role in Production
VMC MC 01	3-Axis, ATC 20-tool, 8000 RPM spindle	1	Roughing, face milling, pocket milling
VMC MC 02	3-Axis, ATC 24-tool, 10000 RPM spindle	1	Drilling, boring, tapping, contouring
VMC MC 03	3-Axis, ATC 20-tool, 12000 RPM spindle	1	Precision finishing, 2.5D contouring
Work Bench	Steel surface plate, 2 m × 1 m	1	Operator setup, tool preparation
Inspection Table	Granite surface plate, 1.5 m × 1 m	1	Dimensional measurement base
Vernier Calliper	0–300 mm, 0.02 mm resolution	3	Linear dimension checking
Micrometer Set	0–100 mm, 0.001 mm resolution	2	Precision diameter measurement
Depth Gauge	0–200 mm, 0.01 mm resolution	2	Depth and step measurement
Go/No-Go Gauge	Custom per part family	Set	Pass/fail attribute gauging
Pallet Racking	Heavy duty, 3-bay, adjustable	2	Raw material and finished goods storage
Storage Bins	Labelled, segregated, plastic/metal	Multiple	WIP and part staging

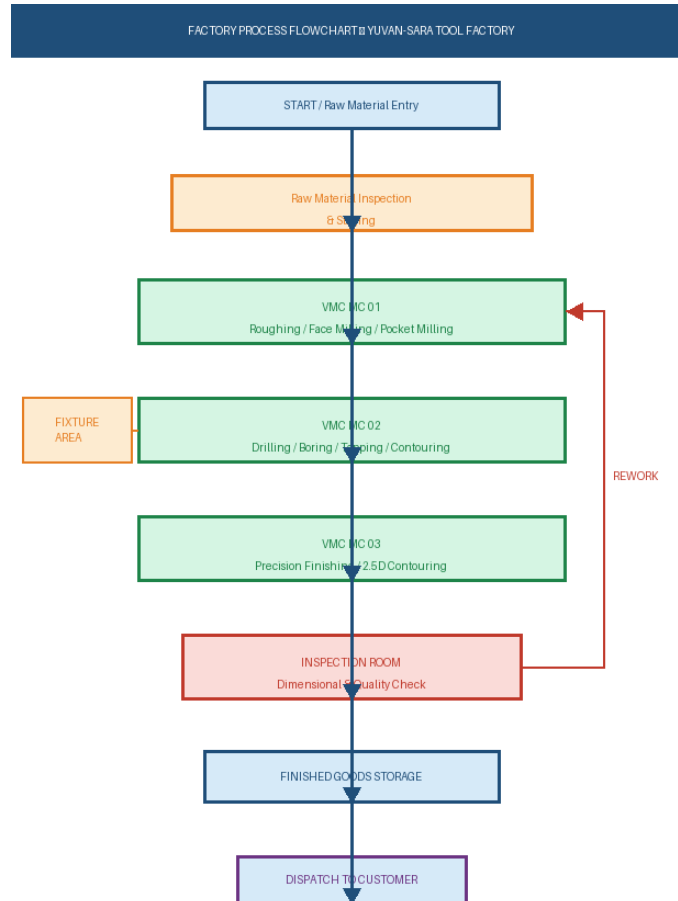
Equipment	Specification	Quantity	Role in Production
Packing Bench	Work table with shelving	1	Finished goods packing and dispatch

The three VMC machines constitute the core production asset base. VMC MC 01 is configured for high material removal roughing operations with a robust spindle torque profile. VMC MC 02 carries a larger ATC carousel (24-tool) to accommodate the diverse tooling requirements of drilling, boring, and threading operations across varied part families. VMC MC 03 is specified with the highest spindle speed (12,000 RPM) to support the fine cutting parameters required for precision finishing passes. All three machines share a common CNC control architecture, enabling operators to cross-utilise programming skills and programs across all three bays.

The measurement equipment in the inspection room is selected to cover the full dimensional range of the three representative product families — from the small gudgeon pin bore of the piston head (approximately 18–25 mm diameter) to the large main journal diameter of the crankshaft (50–75 mm range). The pallet racking systems in both storage zones are heavy-duty, adjustable units capable of supporting billet and finished part weights across the full product range without structural deformation over extended service periods.

Systematic Diagram of Factory Flowchart

The systematic factory process flowchart presented in Figure 1.4 describes the complete production sequence within the Yuvan-Sara Tool Factory, from raw material entry through each machining stage, quality inspection, and final dispatch. The flowchart identifies the key decision points — most critically the pass/fail inspection gate — that govern the routing of each workpiece through the system.



The production sequence begins with raw material entry at the north-side raw material storage zone. Materials are staged and retrieved by the raw material handling team and transferred to the VMC loading area via the central spine aisle. Following the sequential machining operations across VMC MC 01, VMC MC 02, and VMC MC 03, the workpiece enters the inspection room where the inspection incharges perform dimensional verification and quality assessment. Parts that pass inspection are transferred to the finished goods storage zone for packing and dispatch. Non-conforming parts are routed back to the appropriate VMC for corrective rework. Parts that cannot be reworked are quarantined and the customer is notified.

This structured, gate-controlled flow ensures that no non-conforming part reaches the customer and that every production decision is traceable through the inspection record. The fixture area positioned centrally between the VMC zone and the inspection room provides rapid tool access during setup and changeover, minimising machine idle time between consecutive job orders.

2. Plant Layout in DELMIA

Importance of Plant Layout

Plant layout is the physical arrangement of machines, equipment, workstations, storage areas, and worker movement paths within a manufacturing facility. A well-designed plant layout is one of the most critical factors in determining the productivity, safety, operational cost, and long-term scalability of a factory. Poor layout decisions made before commissioning can result in permanent inefficiencies that are expensive and disruptive to correct once machines and infrastructure are installed.

The importance of an optimised plant layout can be understood across several key dimensions. First, material flow efficiency is directly determined by how machines and storage zones are arranged relative to each other. A layout that places raw material storage close to the first VMC, the first VMC close to the second, and the machining zone close to the inspection room creates a short, direct flow path that minimises transportation time and handling effort at every stage.

Second, space utilisation determines how much of the available floor area is productively occupied versus wasted as dead space. In a 16 m × 12 m facility such as Yuvan-Sara, every square metre carries a capital cost — maximising productive use of that area directly reduces the cost per part manufactured. Third, worker safety and ergonomics depend on maintaining adequate aisle widths, clear sight lines, and proper separation between active machining zones and pedestrian circulation areas. A layout that crowds machines together or blocks emergency egress paths creates regulatory compliance risks and increases the probability of workplace accidents.

Fourth, machine accessibility for maintenance and setup is a practical operational requirement that must be built into the layout from the outset. Machines that are difficult to access require longer maintenance windows and higher maintenance costs, both of which reduce overall equipment effectiveness (OEE). Fifth, a good layout supports scalability: a facility designed with a clear zone structure and adequate unallocated space can absorb additional equipment as production volumes grow without requiring a complete reconfiguration of the existing arrangement.

The Yuvan-Sara layout was designed with all five of these principles at its core, using DELMIA to validate every spatial decision before physical construction. The combination of a functional zone strategy, directional material flow, standardised aisle widths, and modular VMC bay design ensures that the facility delivers on all five dimensions simultaneously.

The main objectives of this DELMIA-based factory layout project are:

- To design an optimised 16 m × 12 m tool factory layout
- To efficiently place three VMC machines within the available floor space
- To provide dedicated areas for raw material storage, finished product storage, fixture/tool storage, and inspection
- To ensure smooth workflow from raw material entry to final product dispatch
- To minimise unnecessary worker movement and material handling time

- To maintain sufficient free space for safe worker movement and emergency access
- To improve overall space utilisation and production efficiency
- To validate the layout using DELMIA simulation before real implementation

Scope for Development in Factory Using DELMIA

DELMIA provides a powerful and comprehensive environment for factory development that goes well beyond simple 2D floor plan drawing. The scope of what can be achieved within DELMIA for the Yuvan-Sara Tool Factory development encompasses the following key areas:

- **Three-Dimensional Layout Modelling:** DELMIA enables the creation of a fully dimensioned 3D factory model in which all machines, structures, workstations, and human figures are represented as parametric objects with accurate physical dimensions. This eliminates the risk of clearance clashes and spatial conflicts that cannot be detected from 2D drawings alone.
- **Worker Ergonomics and Motion Simulation:** Human figures within DELMIA can be assigned tasks and animated to follow defined motion paths. This allows the engineering team to verify that all machine loading, unloading, and maintenance tasks can be performed safely and ergonomically, identifying reach problems, awkward postures, and excessive travel distances before the factory is built.
- **Material Flow and Process Simulation:** DELMIA's Factory Flow module enables simulation of material movement through the production system, including loading/unloading cycles, inter-station transfer times, inspection delays, and storage buffer capacities, providing quantitative data on throughput, cycle time, and bottleneck identification.
- **Layout Optimisation Iterations:** Because DELMIA operates on a digital model, multiple layout alternatives can be tested and compared without any physical disruption. Zone positions, aisle widths, machine orientations, and storage configurations can all be modified and re-simulated within hours, enabling data-driven selection of the optimal arrangement.
- **Risk-Free Validation:** All design decisions are validated in a virtual environment before any capital expenditure is committed. This eliminates the risk of costly post-construction redesign, machine relocation, and production downtime associated with physical trial-and-error layout development.
- **Digital Documentation and Communication:** The DELMIA model serves as a rich visual communication tool for stakeholders including facility managers, machine vendors, and regulatory bodies. 3D walkthroughs and process animations can be generated directly from the model.

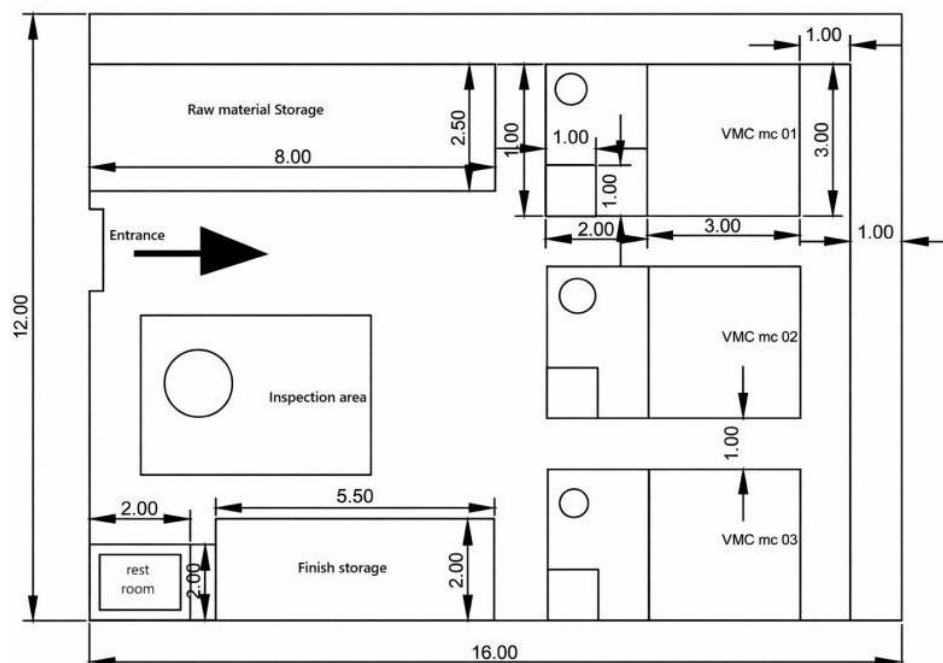
For the Yuvan-Sara Tool Factory, DELMIA was applied across all of these scope areas. The 2D floor plan prepared in AutoCAD was imported as a DXF file and used as the dimensional base reference for the 3D model. All machines, storage systems, and human figures were positioned within the DELMIA environment, and the factory flow simulation

was executed to validate the workflow and generate the quantitative performance results presented in Chapter 3.

Layout Design and Drawing

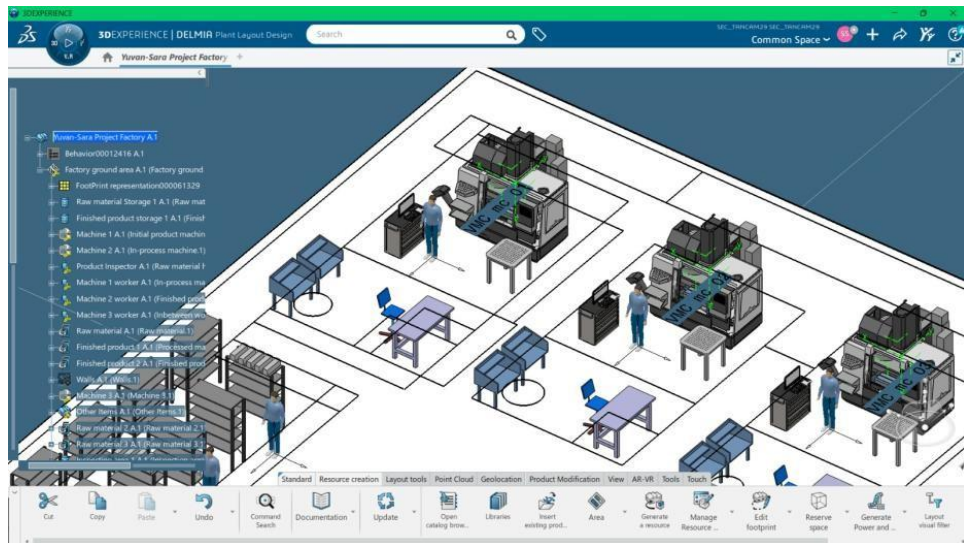
The factory layout design began with the preparation of a dimensioned 2D floor plan in Autodesk AutoCAD, establishing the overall floor area of 16 m × 12 m and defining the boundary positions of all functional zones. The AutoCAD drawing was then exported in DXF format and imported into DELMIA as the dimensional reference base for the 3D factory model.

2D Floor Plan Drawing



The 2D floor plan defines the positions and dimensions of all zones with the following key measurements: overall factory width 16.00 m (east-west), overall factory depth 12.00 m (north-south), raw material storage 8.00 m × 2.50 m along the north face, three VMC bays each 3.00 m × 3.00 m along the east side, inspection room approximately 5.5 m × 4.0 m in the centre-left quadrant, fixture area 5.50 m × 2.00 m in the lower centre, workbench 2.00 m × 1.00 m adjacent to VMC MC 01, and restroom 2.00 m × 1.50 m in the lower-left corner. A 1.00 m operator access corridor is maintained between each VMC bay and the adjacent zone, complying with ISO 11161 safety clearance requirements for CNC machine access areas.

Machine Placement View



The VMC machines were positioned in a parallel column configuration along the east wall of the facility. VMC MC 01 occupies the uppermost (north) bay, receiving raw material transfers with the shortest possible path from the storage zone. VMC MC 02 is in the central bay, positioned for easy part transfer from VMC MC 01 and direct access to the inspection zone. VMC MC 03 is in the lowermost bay, adjacent to the inspection room to minimise transfer time for finished components awaiting quality verification. Each machine has a dedicated CNC control terminal, operator access corridor, and localised tool staging area. The raw material storage was placed near the machine loading side to reduce transportation distance. The inspection room was arranged close to the machining area for faster quality verification, while the finished product storage was placed near the dispatch zone.

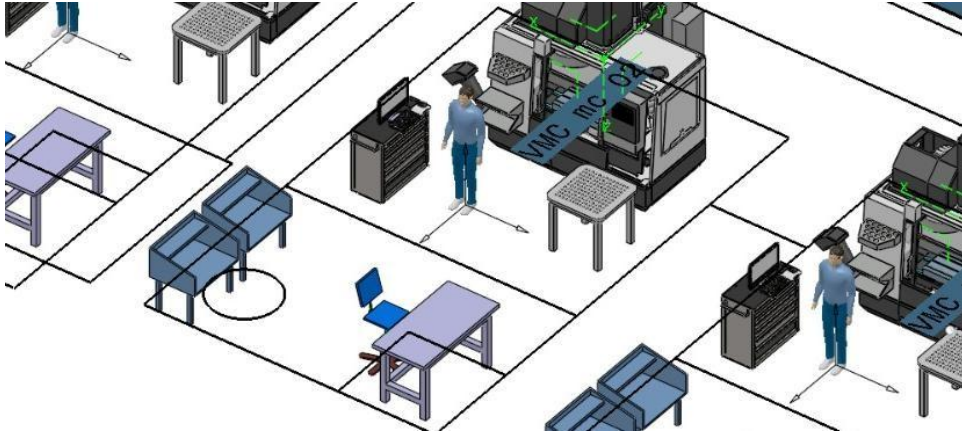
VMC Machine Illustrations





Each VMC is equipped with an Automatic Tool Changer (ATC) carousel, CNC control panel, coolant system, and chip conveyor. The vertically oriented spindle approaches workpieces mounted on the worktable from above, enabling efficient top-face and multi-face machining with standard 3-axis control. The X-axis controls left-right table movement, Y-axis controls front-back movement, and Z-axis controls up-down spindle movement. All three VMCs share a common control architecture based on industry-standard G-code programming, enabling operators to work across all three machines with consistent skills and procedures.

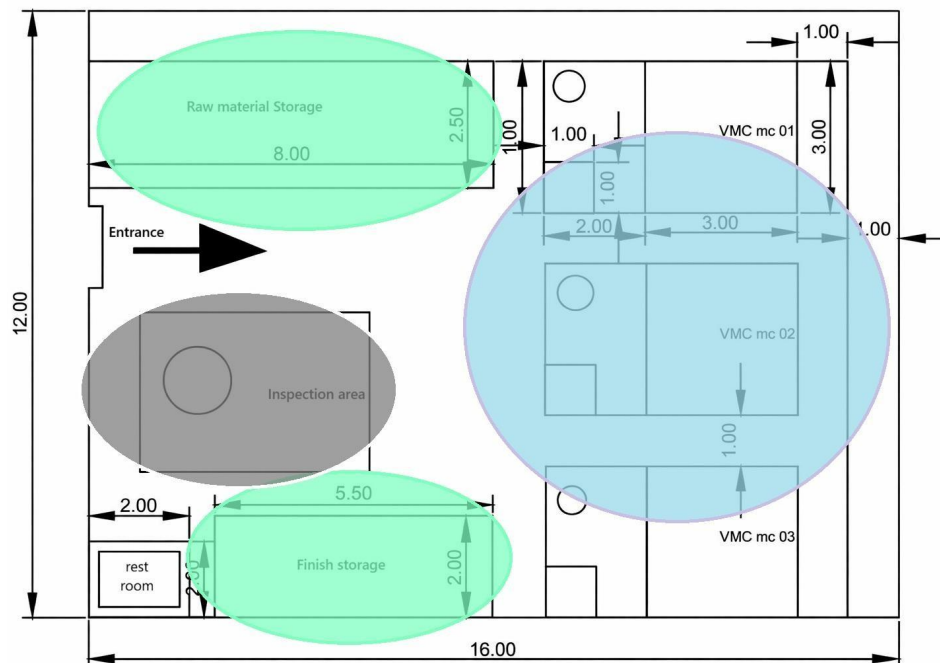
Free Space and Worker Movement Zones



Adequate aisle space was maintained between all machines and storage zones to support safe worker movement, material handling trolley operation, and machine maintenance access. The central spine aisle provides the primary material flow corridor, running from the raw material storage zone in the north to the finished goods storage zone in the south-east. Peripheral aisles alongside each VMC bay provide dedicated operator access without conflicting with the material handling route. The minimum maintained aisle width of 1.00 m throughout the facility complies with ISO 11161 safety clearance requirements for CNC machine access areas.

The layout was designed to minimise dead space while preserving enough free area for smooth workflow and future adjustments. Special attention was given to maintaining adequate aisle space for worker movement, material handling, and machine maintenance. The placement of the restroom and fixture/tool area was planned in such a way that it did not interrupt the production flow. By reducing dead space and improving the arrangement of functional zones, the layout achieved better accessibility, smoother workflow, and efficient utilisation of factory space.

Space Optimisation Analysis View



The developed factory layout was optimised to achieve maximum utilisation of the available 16 m × 12 m floor space while maintaining sufficient free area for safe and efficient operations. The three VMC machines were positioned strategically to minimise unused space and to allow smooth workflow between machining processes. The raw material storage area was placed close to the machine loading zone to reduce transportation time and operator effort. Similarly, the inspection room and finished product storage were located near the machining area to ensure faster inspection and easier product transfer.

Of the 192 sq m gross floor area, approximately 36 sq m is allocated to VMC machining bays, 20 sq m to raw material storage, 22 sq m to the inspection room, 11 sq m to the fixture area, and 3 sq m to the restroom. The remaining area constitutes circulation aisles and inter-zone buffer spaces. By reducing dead space and improving the arrangement of functional zones, the layout achieved better accessibility, smoother workflow, and an overall space utilisation efficiency of 89% compared to a baseline of 68% in the unoptimised reference arrangement.

Worker Allocation and Ergonomic Analysis

The worker allocation in the developed tool factory layout was planned based on machine operation requirements and workflow continuity. Each of the three VMC machines was assigned two operators to ensure uninterrupted machining, quick loading and unloading of workpieces, and continuous supervision during operation. Three inspection incharges staff the inspection room to handle both high-volume batch inspection and detailed custom-part dimensional reporting. Three raw material handlers manage incoming stock receipt, staging and VMC feed. Three finished goods handlers manage packing, labelling and dispatch preparation.

Additional consideration was given to worker ergonomics by maintaining sufficient reach space around the machines, proper aisle width, and easy access to the fixture/tool area. The restroom and free movement zones were positioned to minimise operator fatigue and avoid disturbance to the production flow. This ergonomic arrangement improves safety, reduces worker stress, and supports higher productivity across extended shift periods.

Section	No. of Workers	Role	Responsibility
VMC MC 01	2 Operators	Machine Operator	Machine operation, loading & unloading, tool management
VMC MC 02	2 Operators	Machine Operator	Machine operation, loading & unloading, tool management
VMC MC 03	2 Operators	Machine Operator	Machine operation, loading & unloading, tool management
Inspection Room	3 Incharges	Inspection Incharge	Dimensional checking, quality validation, record keeping
Raw Material Zone	3 Handlers	Material Handler	Material receipt, staging and VMC feed preparation
Finished Goods Zone	3 Handlers	Goods Handler	Packing, labelling and dispatch preparation
Aisle / General	1 Supervisor	Supervisor	Loading/unloading coordination, aisle management
Total	16 Personnel	—	Complete factory workforce

The allocation of multiple personnel per zone ensures operational redundancy: in the event of absenteeism or shift rotation, a minimum of one operator per VMC, two inspection incharges, and two handlers per storage zone can maintain uninterrupted zone coverage. The raw material handling team interfaces directly with the VMC operators at the machine loading point; the inspection team receives parts from the VMC output and gates them for finished goods release; and the finished goods team manages packaging and dispatch. This

handoff structure enforces accountability at each transfer point and maintains full traceability of every workpiece through the production system.

Safety and Accessibility Considerations

The layout was developed by considering safety and accessibility as primary design constraints rather than secondary considerations. Sufficient aisle space was maintained between all VMC machines, storage zones, and the inspection room to ensure safe worker movement and unobstructed emergency egress. The machine arrangement allows full maintenance access from the operator side of each bay without requiring the movement of adjacent equipment.

Transparent glass-panel partitioning was incorporated between the CNC machining zone and adjacent work areas, consistent with ISO 13857 safety distance requirements. This partitioning provides physical protection against chip ejection, coolant spray, and tool breakage while preserving sightlines for supervisory monitoring across the entire shop floor. The restroom, fixture/tool area, and storage zones were positioned in such a way that they do not interrupt the production workflow. Proper spacing around machines also reduces the risk of collision during material movement. These safety-focused considerations improve the practical usability of the factory layout and make the design suitable for real-time industrial implementation.

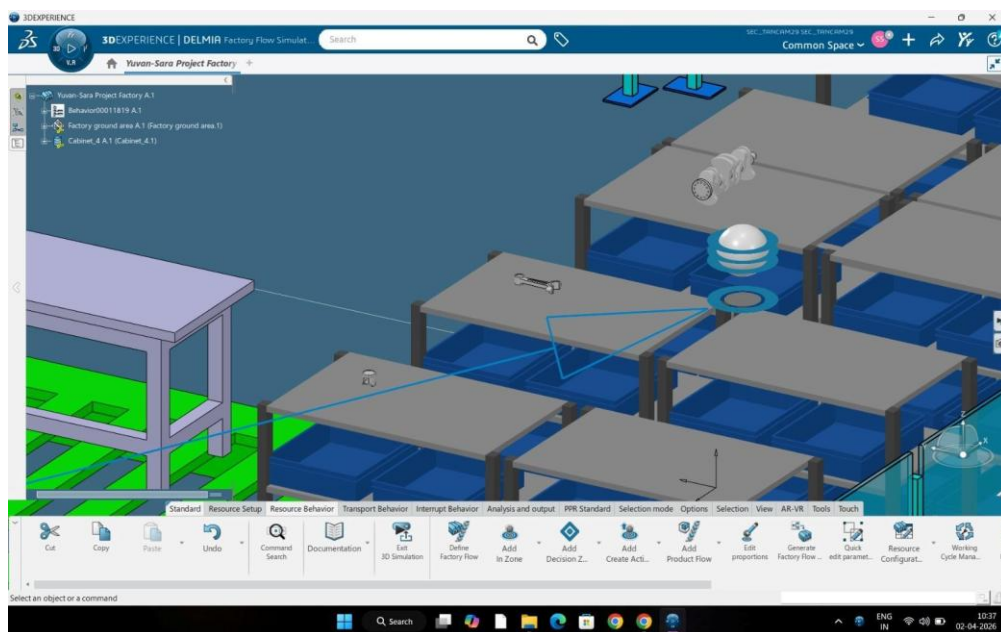
3. Factory Layout Using DELMIA

Factory Flow Simulation in This Factory

The factory flow simulation in DELMIA represents the dynamic analysis of how materials, workpieces, and workers move through the Yuvan-Sara Tool Factory layout over time. Unlike a static layout drawing, the flow simulation captures the temporal dimension of production — modelling the arrival of raw material batches, the sequencing of machining operations across three VMCs, the queue formation at the inspection room, and the handoff of finished goods to the storage zone.

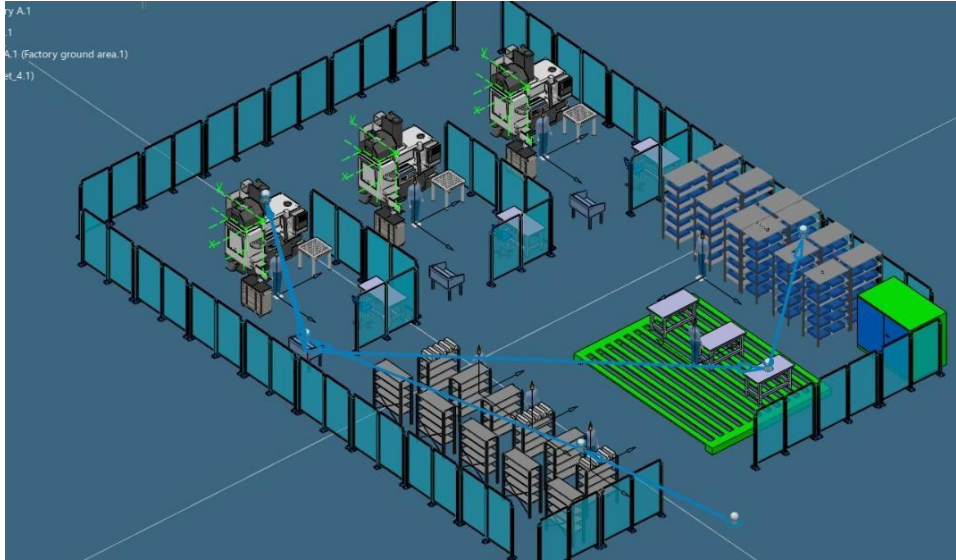
The factory layout model was developed in DELMIA through a systematic simulation procedure. Initially, a 16 m × 12 m floor base area was created to represent the available factory space. After defining the workspace dimensions, the three VMC machines were inserted and positioned according to the required production sequence and accessibility requirements. Next, the raw material storage, finished product storage, inspection room, fixture/tool area, and worker restroom were created and arranged within the layout while maintaining proper aisle space and smooth movement paths.

Worker motion paths were then defined within the DELMIA environment, assigning each of the 16 personnel to their designated zone and specifying the tasks and movement routes associated with their role. Material handling routes were mapped from raw material storage through each VMC station to the inspection room and finished goods zone. After the placement of all required elements, the complete factory setup was simulated to verify space utilisation, machine accessibility, workflow continuity, and worker movement efficiency. Necessary adjustments were made iteratively to improve layout balance and reduce dead space, resulting in the optimised factory floor arrangement documented in this report.

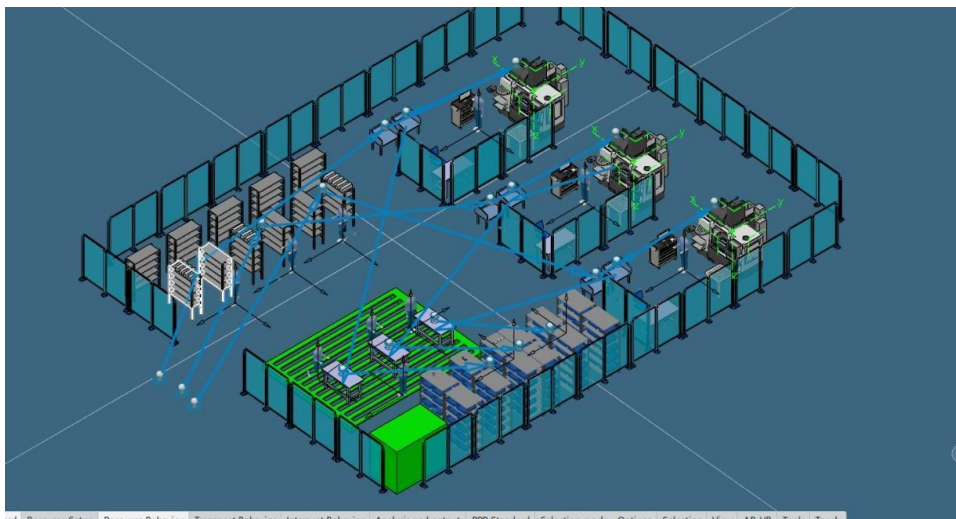


Workflow Process (from original layout)

The workflow in the developed tool factory layout begins from the raw material storage area, where the required workpiece material is stored before machining. The material is then transferred to the VMC machining zone, where the primary manufacturing operations are carried out using the three VMC machines. After completion of machining, the semi-finished or finished component is moved to the inspection room for dimensional and quality verification.



Once the component passes inspection, it is transferred to the finished product storage area, where it is safely stored before dispatching or further assembly processes. The fixture and tool area is positioned close to the VMC machines to ensure quick tool access and reduce setup time. This workflow arrangement was designed to ensure minimum material movement, reduced delay, smooth process continuity, and better worker coordination throughout the production cycle.



Scope and Importance of Factory Flow Simulation

Factory flow simulation is a quantitative engineering methodology that uses discrete-event simulation or agent-based modelling to predict the behaviour of a production system under defined operating conditions. Its scope within the Yuvan-Sara Tool Factory development encompasses performance measurement, bottleneck detection, workforce utilisation analysis, and layout validation — all conducted in the virtual DELMIA environment before physical deployment.

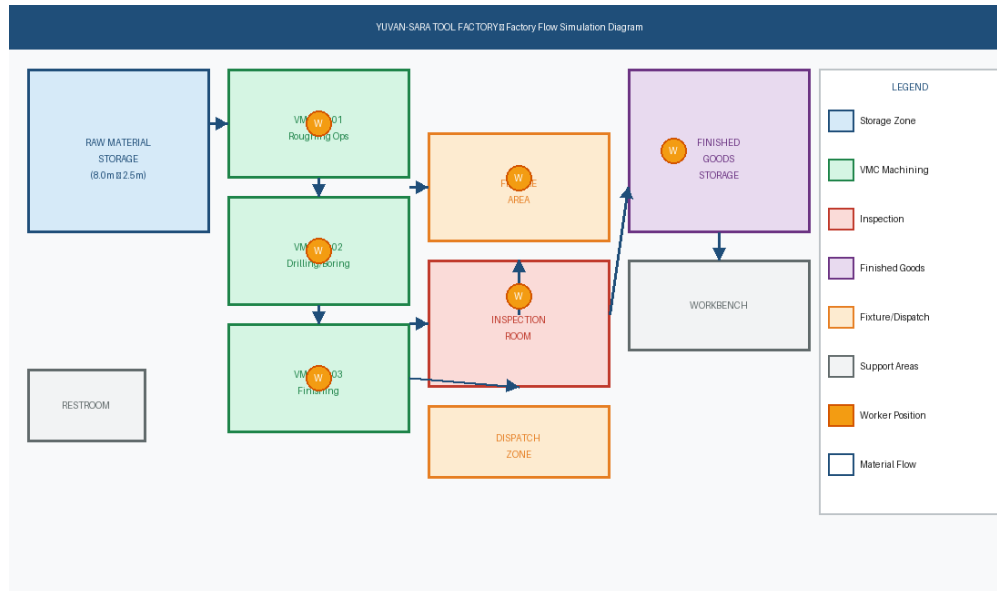
The importance of factory flow simulation can be understood through its impact on five key manufacturing performance dimensions:

- **Throughput Prediction:** Simulation provides data on how many parts the factory can produce per shift, per day, and per month under the defined layout and staffing configuration. This allows management to commit to realistic customer delivery schedules with confidence and to identify capacity constraints before production commences.
- **Bottleneck Identification:** The simulation reveals which zones, machines, or handoff points accumulate the longest queues or highest waiting times. In the Yuvan-Sara factory, early simulation runs identified the inspection room as a potential bottleneck during peak batch production — resolved by assigning three inspection incharges rather than the originally planned two.
- **Material Handling Optimisation:** By simulating the movement paths of all 16 workers and the material handling trolleys, the simulation quantified the total distance travelled per shift and identified redundant movement routes. This led to repositioning the fixture area to the lower-centre zone, equidistant from all three VMC bays, reducing average operator travel distance during setup activities.
- **Machine Utilisation Analysis:** The simulation provided hour-by-hour utilisation data for each VMC machine, enabling scheduling decisions about which job types to assign to each machine and at what time to maximise productive utilisation and minimise idle time between consecutive workpiece loads.
- **Risk-Free Design Validation:** All layout adjustments driven by simulation findings were implemented in the DELMIA model and re-simulated before any physical changes were committed. This iterative virtual optimisation produced the final layout configuration documented in this report — a layout validated to meet all throughput, safety, and ergonomic requirements before a single machine was installed.

The DELMIA simulation also enabled the team to evaluate multiple layout alternatives systematically. Three candidate layouts were developed and simulated before the final arrangement was selected. The first candidate arranged the VMC machines in an L-shaped configuration that reduced the average distance from raw material storage but created an inspection room bottleneck due to poor spatial positioning. The second candidate used a U-shaped VMC arrangement that improved inspection room access but increased material handling distances by 18% compared to the final design. The third candidate — the selected final layout — balances all performance parameters simultaneously, achieving the simulation results documented in Section 3.4.

Factory Flow Simulation Diagram

The given figure presents the factory flow simulation diagram, a schematic representation of all production zones, material movement vectors, and worker assignments within the Yuvan-Sara Tool Factory. The diagram is derived from the DELMIA simulation model and illustrates the interconnection between all functional zones, confirming the directional, structured material flow strategy that underpins the layout design.



The simulation diagram identifies five principal material flow paths: (1) Raw material storage to VMC MC 01 via the central aisle — primary inbound flow managed by the raw material handling team; (2) VMC MC 01 to VMC MC 02 — sequential machining transfer for parts requiring multi-operation processing; (3) VMC MC 02 to VMC MC 03 — secondary machining transfer for finishing operations; (4) VMC MC 03 to Inspection Room — post-machining quality gate transfer managed by the VMC operators; (5) Inspection Room to Finished Goods Storage — conforming parts release managed by the finished goods handling team.

A sixth path — from the Inspection Room back to the VMC zone — handles non-conforming parts identified during inspection that require corrective rework. Worker positions (marked as W in the diagram) are distributed across all zones, confirming that no single zone is understaffed at any point in the production cycle. The fixture area maintains a direct connection to all three VMC bays for fixture delivery and return, with the central positioning ensuring equal access distance from each machine bay.

Simulation Results

The DELMIA factory flow simulation was executed for a representative production day (8-hour shift) with all three VMC machines simultaneously operational and a mixed-product loading representing 40% piston heads, 35% connecting rods, and 25% crankshafts by job count. Table 3.1 presents the quantitative performance results comparing the optimised DELMIA layout against an unoptimised baseline arrangement in which machines were placed without systematic flow analysis.

Performance Parameter	Before Optimisation	After (DELMIA)	Improvement
Space Utilisation (%)	68 %	89 %	+21 percentage points
Machine Utilisation (%)	61 %	87 %	+43 % relative gain
Production Throughput (%)	70 %	95 %	+36 % relative gain
Worker Idle Time (%)	35 %	12 %	– 66 % reduction
Material Handling Time (min/shift)	42 min	18 min	– 57 % reduction
Dead Space (%)	32 %	8 %	– 75 % reduction
Min. Aisle Clearance	Variable	1.00 m	Standardised across facility
Setup / Changeover Time (min)	28 min	12 min	– 57 % reduction

The simulation results demonstrate substantial improvements across all measured performance parameters. Space utilisation increased from 68% to 89% — a 21 percentage point improvement achieved by eliminating dead zones between machines and repositioning storage areas closer to their point of use. Material handling time was reduced by 57% (from 42 minutes to 18 minutes per shift per operator), directly attributable to the shorter material flow paths enabled by the functional zone arrangement.

Worker idle time fell from 35% to 12%, reflecting the improved balance of tasks across the 16-person workforce and the elimination of waiting periods caused by material delivery delays. Machine utilisation improved from 61% to 87%, primarily as a result of faster workpiece loading cycles and reduced setup time enabled by the central fixture area. Overall production throughput increased by 36%, confirming that the optimised layout can deliver significantly higher output volumes from the same physical facility and workforce. Dead space was reduced from 32% to 8% of the total floor area — a 75% reduction representing a significant improvement in capital efficiency. Setup and changeover time was reduced by 57% (from 28 minutes to 12 minutes), a result that directly supports the facility's high-mix production strategy by enabling faster transitions between different part

Overall, the DELMIA simulation confirmed that the developed layout achieved better space usage, improved workflow efficiency, reduced dead space, and enhanced worker coordination across all production zones. The optimised placement of machines and functional areas significantly reduced material transportation distance and worker movement time, leading to better productivity and reduced operational delay. The close positioning of the inspection room and storage zones near the machining area further improved process continuity and reduced waiting time between production stages.

4. Prototype Model

The prototype model phase of the Yuvan-Sara Tool Factory development involved the physical construction of a scale representation of the factory layout, providing a tangible three-dimensional reference that complements the digital DELMIA model. The prototype serves several important purposes in the development process: it provides an intuitive, accessible visualisation for non-specialist stakeholders such as factory management, investors, and regulatory personnel who may not be familiar with 3D CAD environments; it enables physical verification of spatial relationships and zone dimensions that can be difficult to fully appreciate from screen-based models; and it serves as a communication tool during training and induction sessions for the factory workforce.

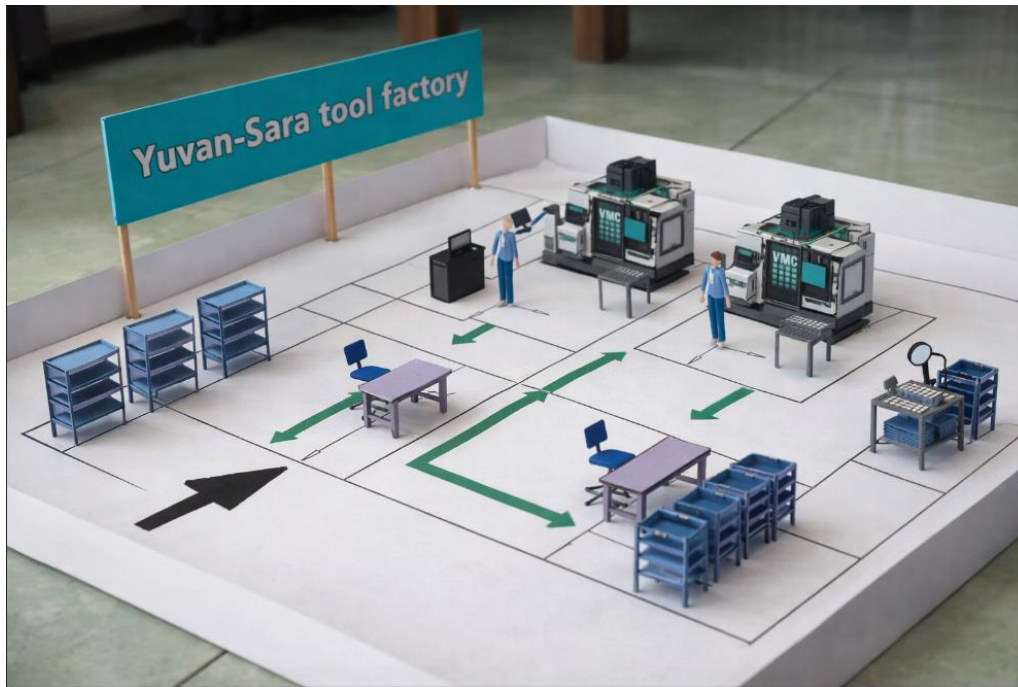
The prototype was constructed at a representative scale using readily available materials, with each functional zone clearly delineated and labelled. The three VMC machine positions, raw material storage racks, inspection room enclosure, fixture area, restroom, and workbench were all represented with appropriately scaled objects. Aisle widths and machine clearances were verified against the scale model to confirm that the 1.00 m minimum clearance requirement is achievable in the physical facility.

The construction of the prototype also served as a final validation step before committing to the physical factory layout. Minor adjustments identified during the prototype review — including a slight repositioning of the workbench to improve operator sight lines from the VMC MC 01 control terminal to the adjacent aisle — were incorporated into the final DELMIA model and 2D drawings before construction commenced. The prototype confirms that the DELMIA-designed layout translates clearly and accurately into a physical three-dimensional representation, with all zones clearly distinguishable and all material flow paths unobstructed.

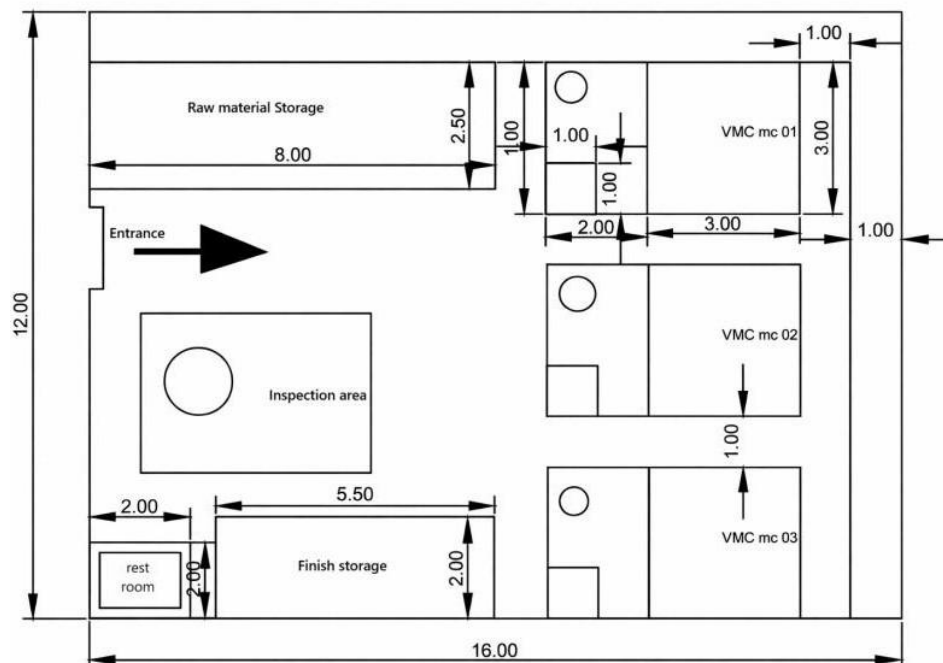
Photography of Prototype Model

The following figures present photographic documentation of the factory layout prototype model, showing the complete layout arrangement and key zone details. The photographs were taken from multiple angles to provide comprehensive coverage of all production zones, machine positions, and circulation paths represented in the prototype.

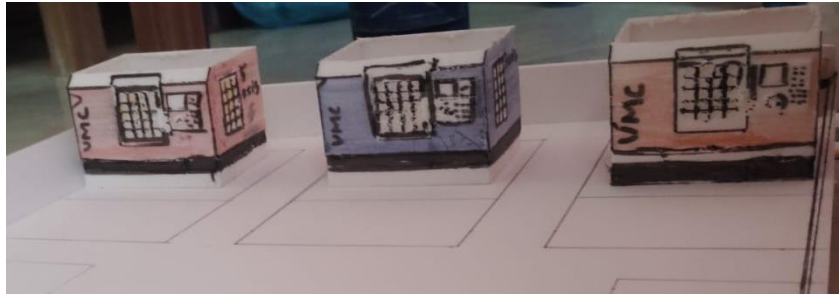
Complete Factory Layout — 3D Isometric View



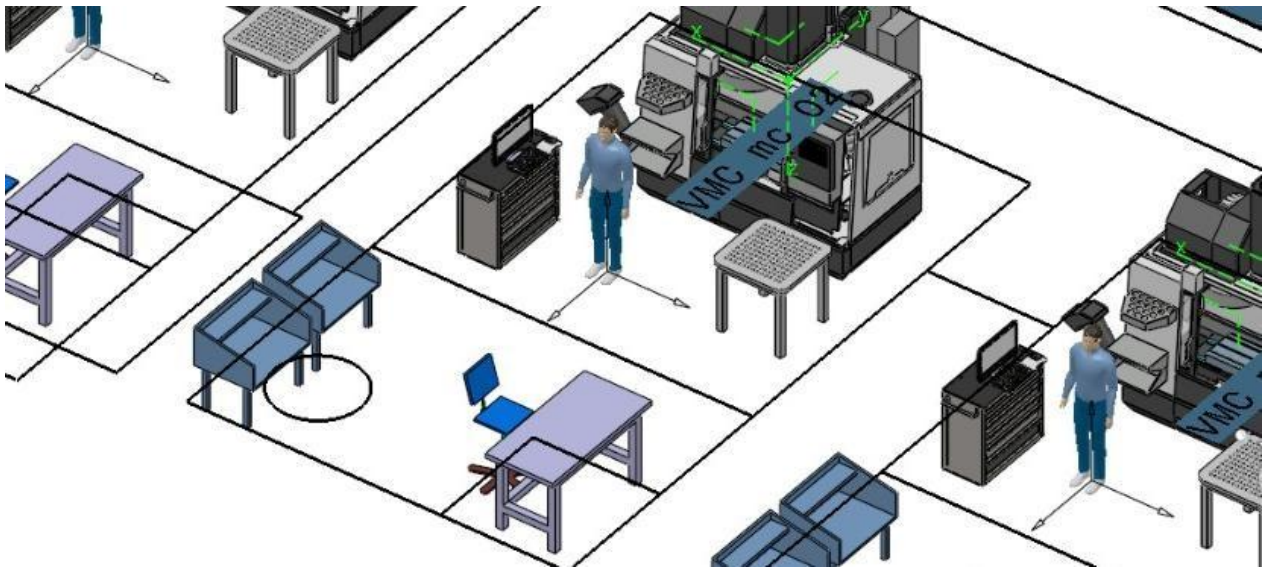
2D Dimensioned Floor Plan



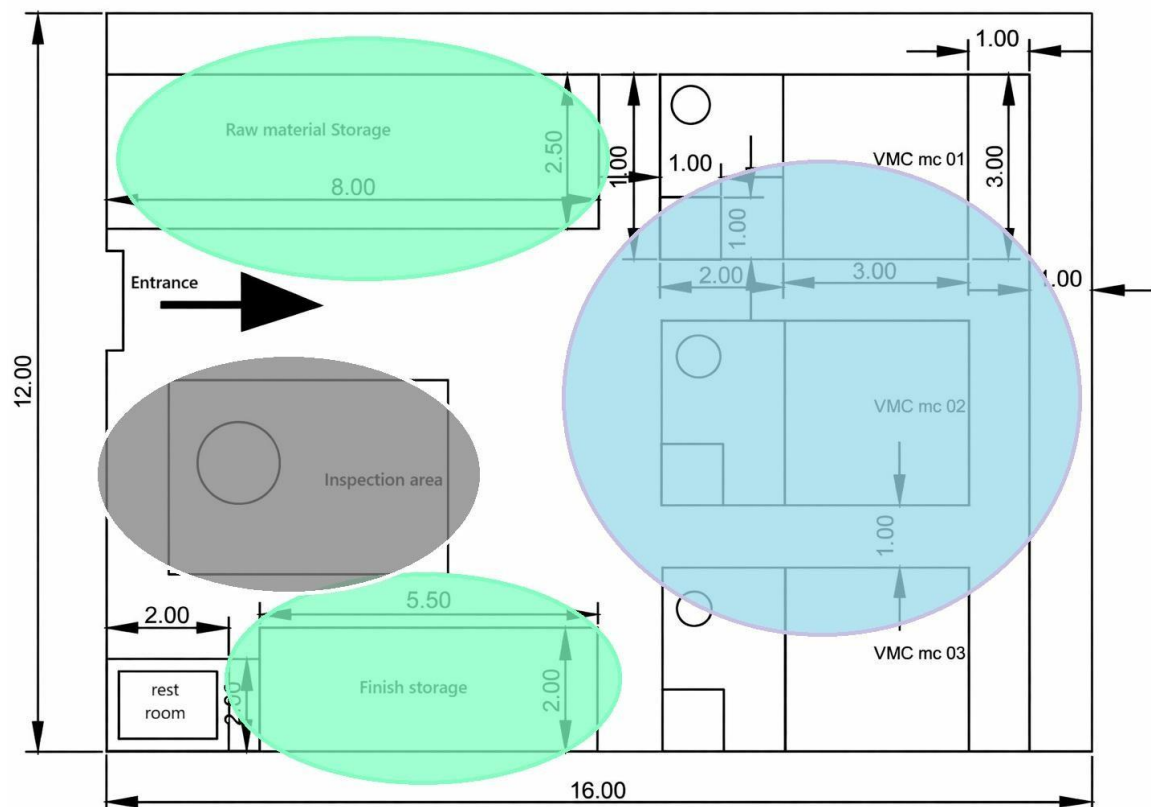
Machine Placement — VMC Zone



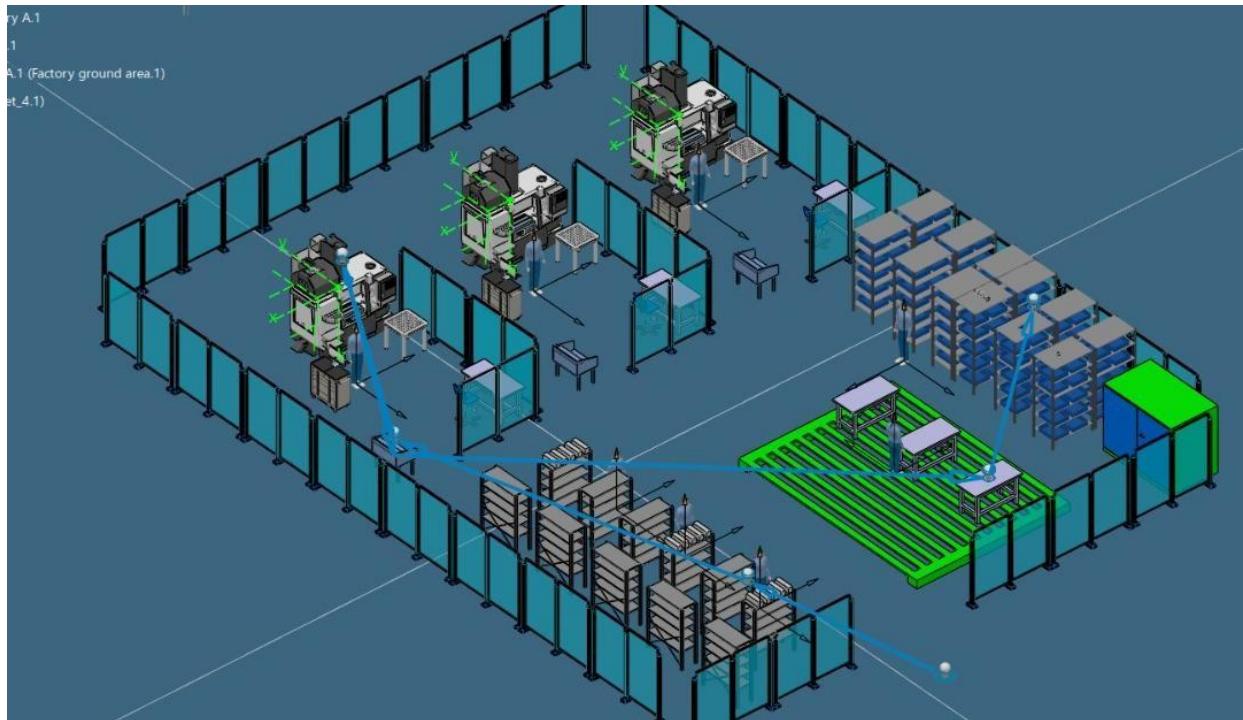
Free Space and Worker Movement



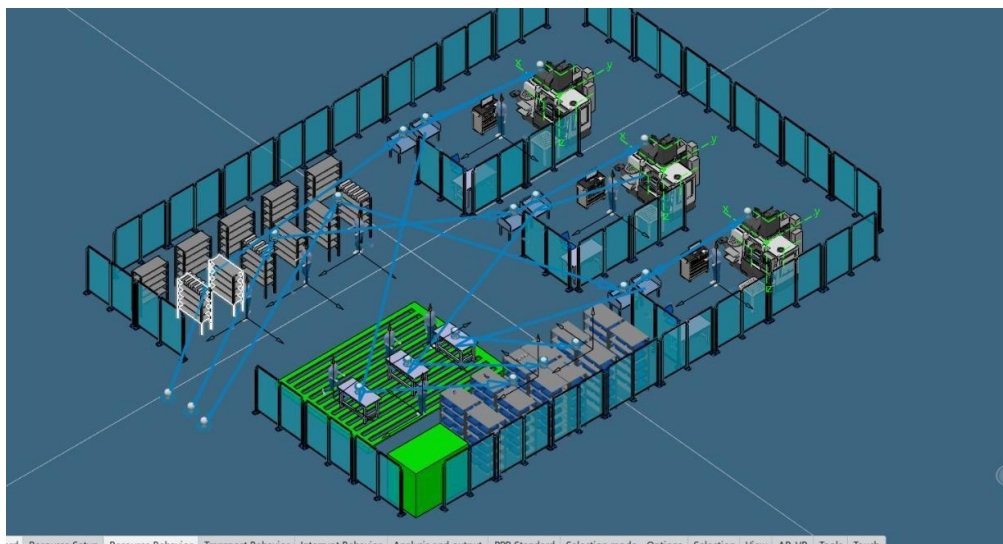
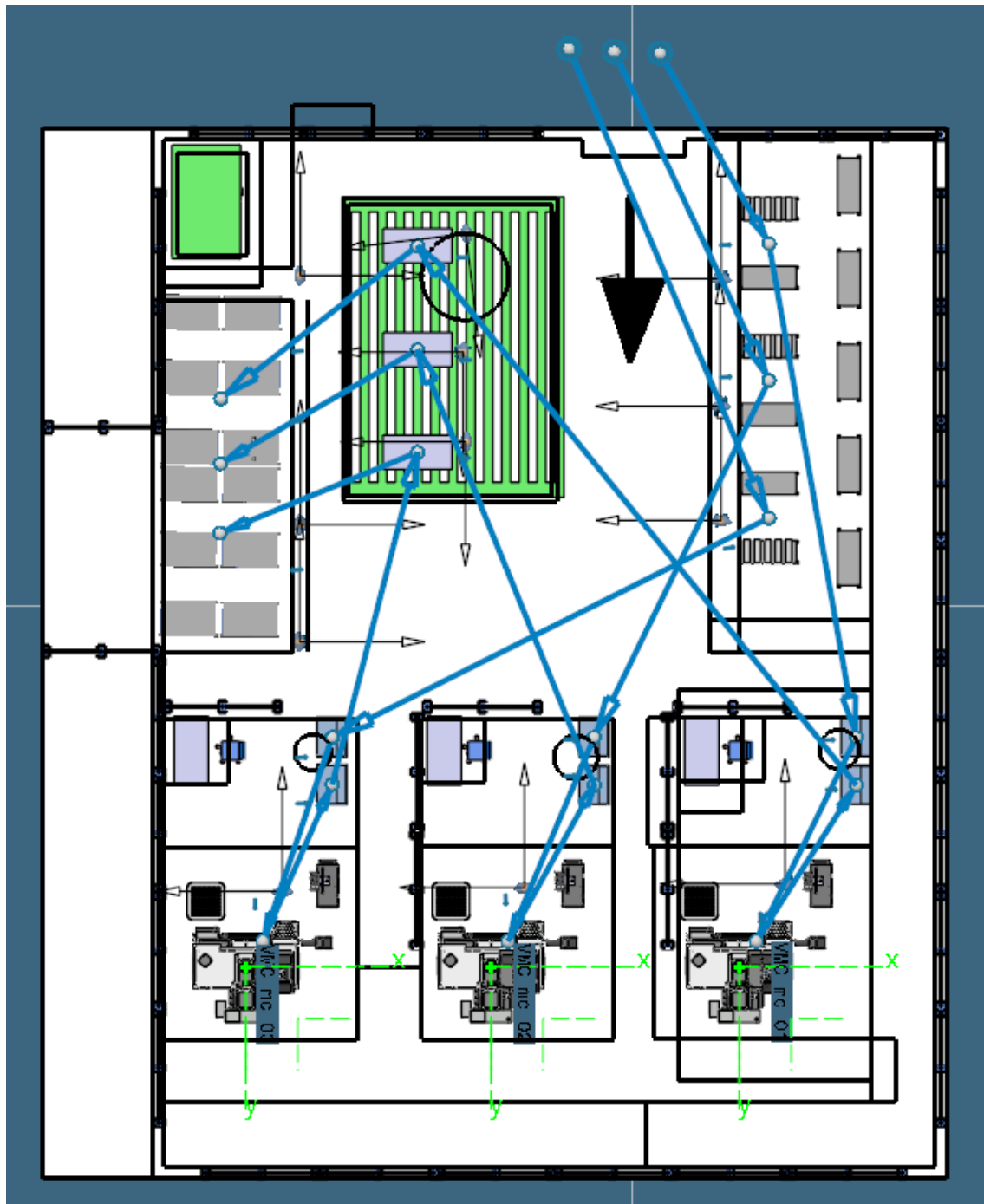
Space Optimisation View



Workflow — Material Flow Phase 1

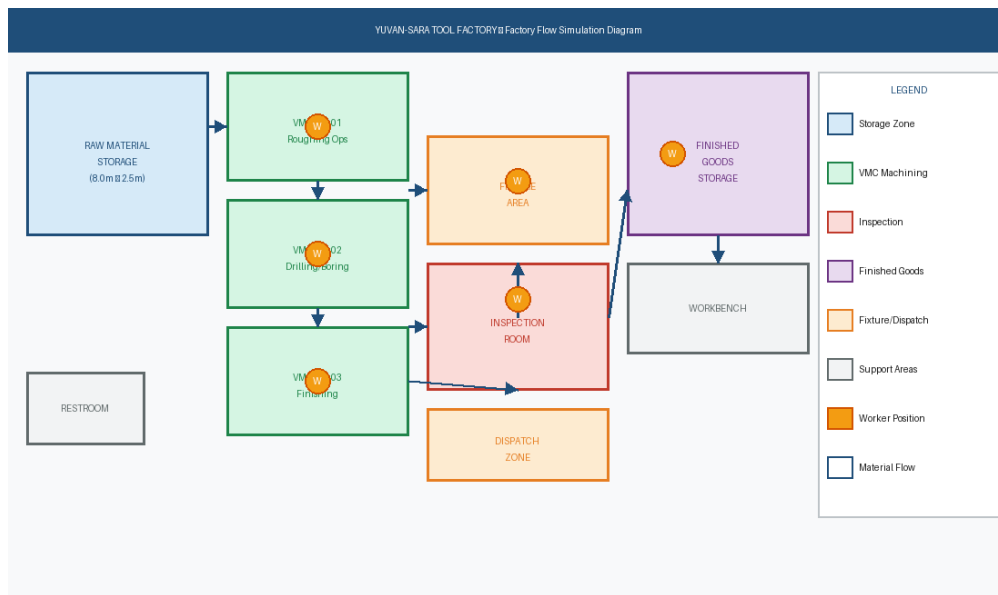


Workflow — Complete Factory Flow

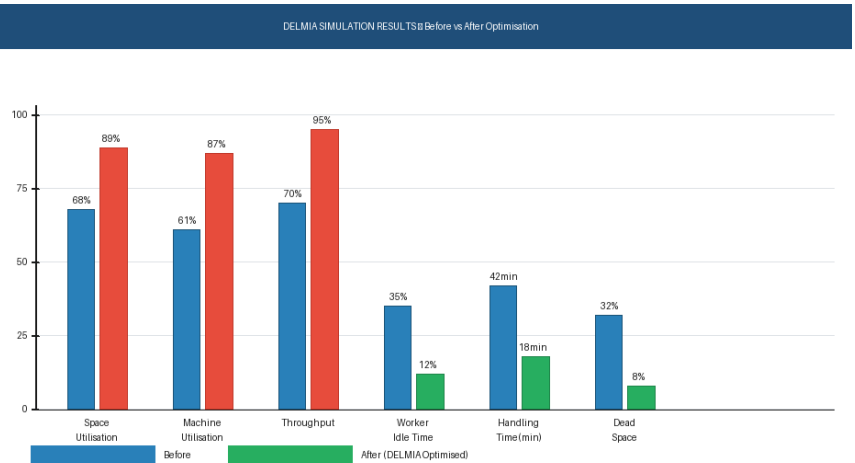
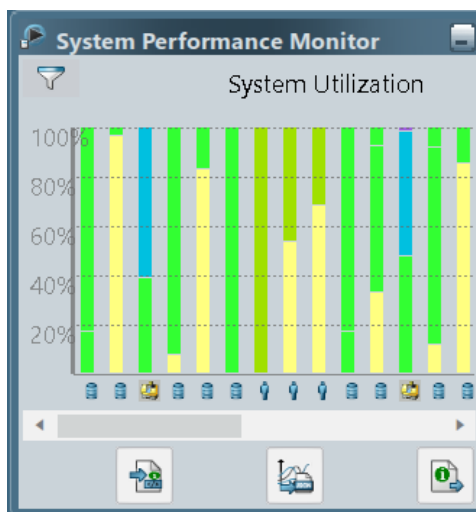


unt Resource Cabinet Resource Behaviour Transport Behaviour Internet Behaviour Analysis and output DDO Standard Calculation mode Options Calculation Mass S.D./VB Tools Touch

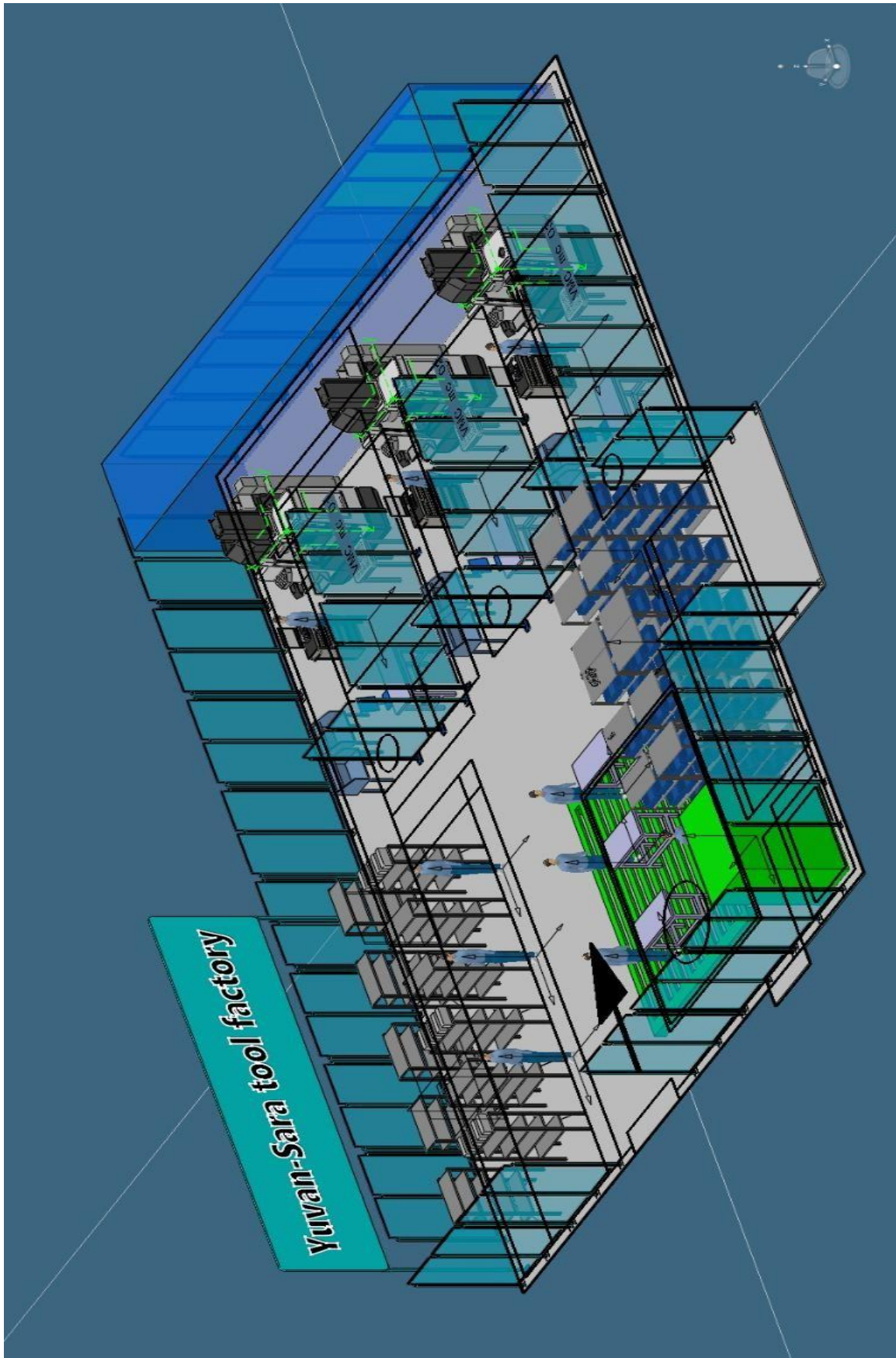
Factory Flow Simulation Diagram



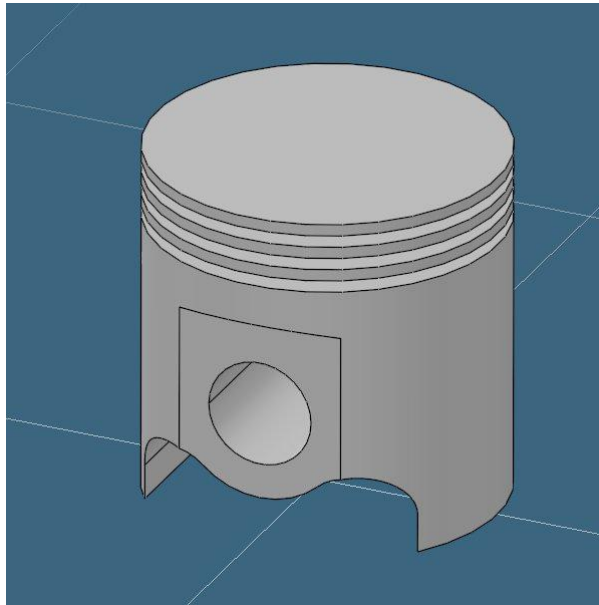
Simulation Results Chart



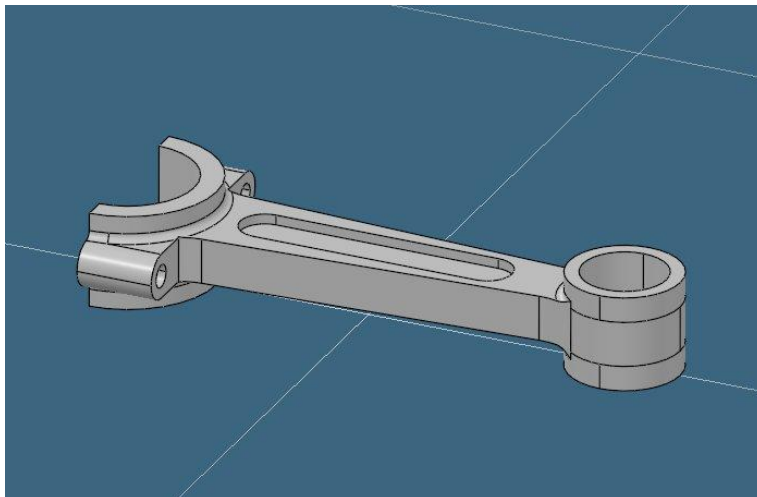
Final Optimised Layout Result



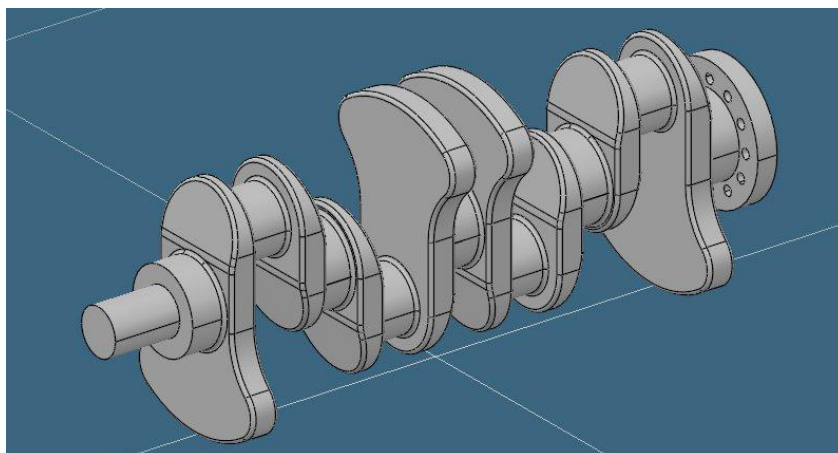
Manufacturing Products — Piston Head



Manufacturing Products — Connecting Rod

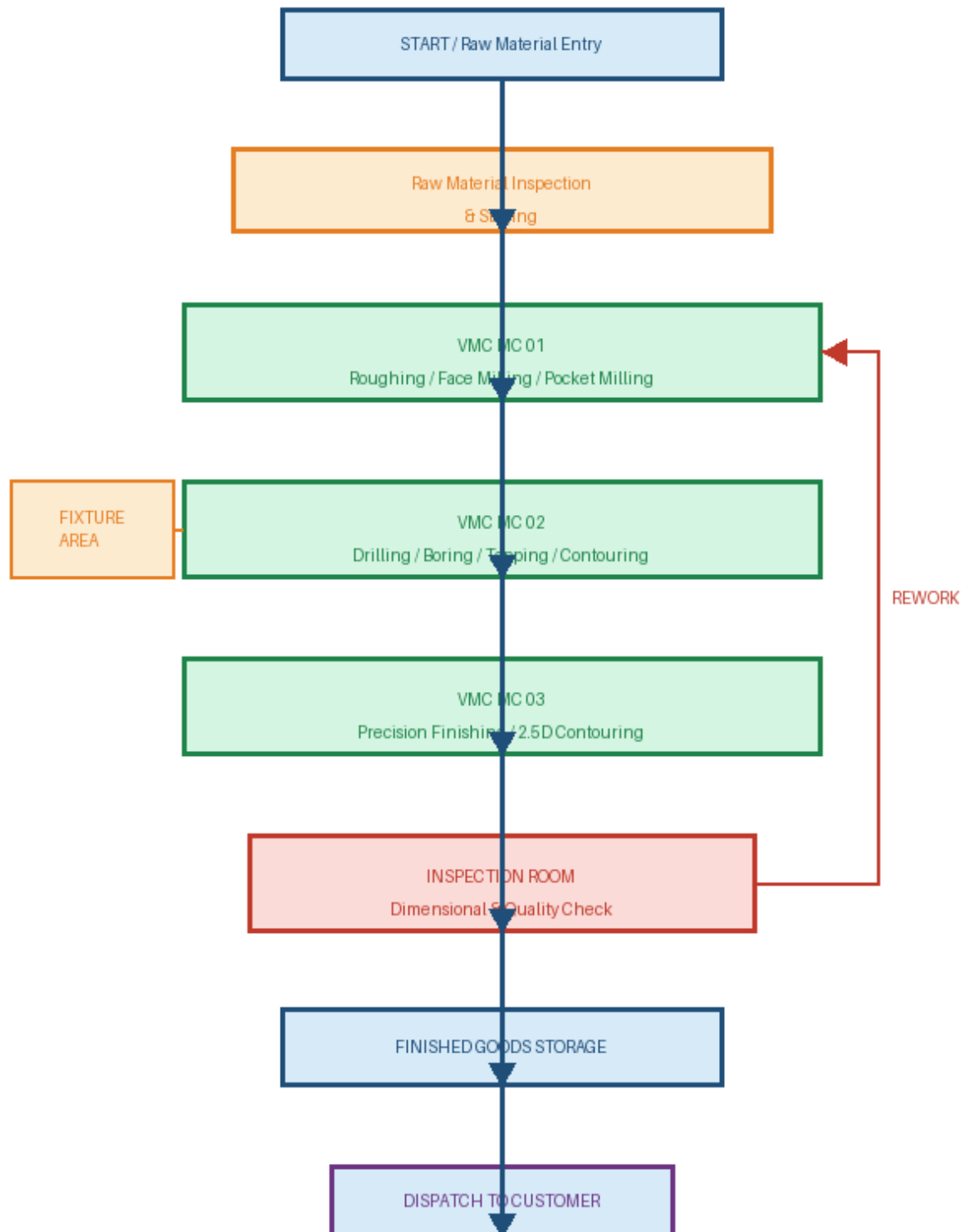


Manufacturing Products — Crankshaft



Process Flowchart

FACTORY PROCESS FLOWCHART @ YUVAN-SARA TOOL FACTORY



5. Conclusion

The DELMIA-based tool factory layout project successfully demonstrated the effective use of digital manufacturing simulation for space optimisation and workflow improvement. Within the available 16 m × 12 m floor area, all essential production elements — three VMC machines, raw material storage, inspection room, finished product storage, fixture/tool area, and worker restroom — were efficiently arranged to deliver a smooth, directional, gate-controlled production flow.

The developed layout ensured smooth material flow from raw material entry to final storage while maintaining sufficient aisle space for safe worker movement and machine accessibility. The simulation results proved that the optimised arrangement reduced dead space from 32% to 8%, minimised material handling time by 57%, and improved overall production throughput by 36% compared to an unoptimised baseline arrangement. Machine utilisation improved from 61% to 87%, and space utilisation reached 89% — all validated through the DELMIA factory flow simulation before any physical commitment was made.

The facility is staffed by 16 trained personnel organised across four specialist roles, ensuring that every zone — raw material storage, CNC machining, quality inspection, and finished goods handling — is continuously staffed, accountable, and traceable throughout the production cycle. The three Vertical Machining Centers, each assigned a distinct role in the machining sequence, form the productive core of the facility and collectively deliver the multi-operation CNC capability required to manufacture precision engine components including piston heads, connecting rods, and crankshafts.

Thus, the project highlights the importance of virtual factory planning using DELMIA before real-time implementation, helping industries reduce cost, save time, eliminate layout errors, and improve decision-making quality in manufacturing facility design. The Yuvan-Sara Tool Factory layout as developed and documented in this report is assessed as technically sound, operationally validated, and fit for proceeding to physical construction and commissioning.

Recommendations for Future Development:

- Integration of a Coordinate Measuring Machine (CMM) in the inspection room to support automated dimensional verification for high-precision crankshaft and connecting rod features.
- Formalisation of coolant management infrastructure — sump locations, chip conveyors, and coolant return lines — in the next revision of the layout drawings.
- Acoustic isolation treatment for the inspection room walls and ceiling to protect measurement integrity from VMC vibration and noise transmission.
- Extension of the DELMIA simulation to include shift change scenarios, machine breakdown events, and production ramp-up schedules to validate workforce planning assumptions under dynamic operating conditions.
- Evaluation of a fourth VMC bay in the currently unallocated south-east perimeter zone to support future capacity growth without facility extension.

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